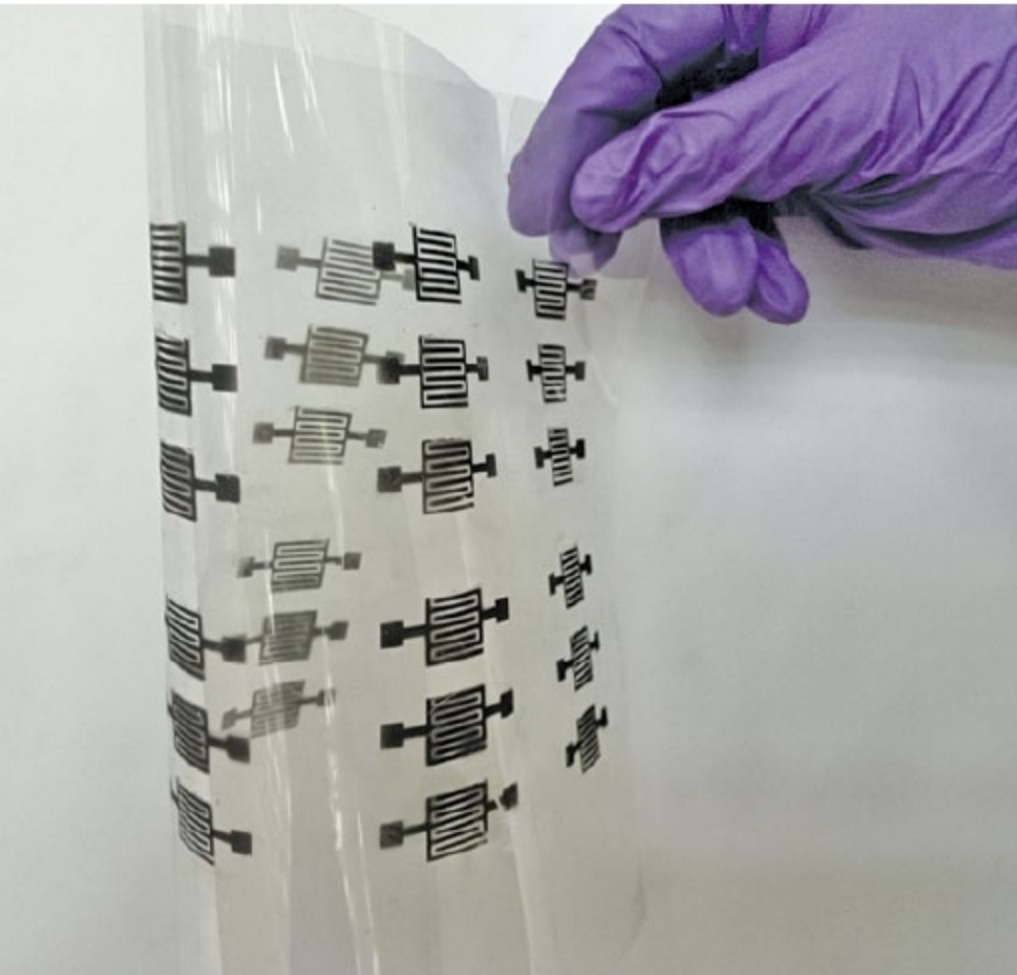


Flexible supercapacitor developed using simple spray coating method



Bending the supercapacitor-printed sheet does not affect performance, making it useful for flexible energy storage applications (Credit: Indian Institute of Science)

Supercapacitors could one day replace batteries used in electric cars, cellphones or laptops, because these charge quickly, and work at almost 100 per cent efficiency. A team at Indian

Institute of Science (IISc) has created a compact and flexible supercapacitor by using a simple spray coating technique to deposit alternating layers of hybrid nanocomposites on a bendable plastic sheet. The layer-by-layer patterning increases the surface area and boosts the movement of charges, making the device more efficient than existing supercapacitors.

Abha Misra, senior author and associate professor at Department of Instrumentation and Applied Physics, IISc, said, "We can actually print these supercapacitors anywhere, on any substrate. Thus, these can easily be mounted on any surface, just like a simple spray on the walls."

Unlike a battery that uses chemical reactions, a supercapacitor uses static electricity to store charge. It has two electrodes dipped in an electrolyte and separated by a thin insulator. When the electrodes are charged, an electric field is created between these, which allows energy to be stored. The larger the surface area of the electrodes, the greater is the charge that can be stored.

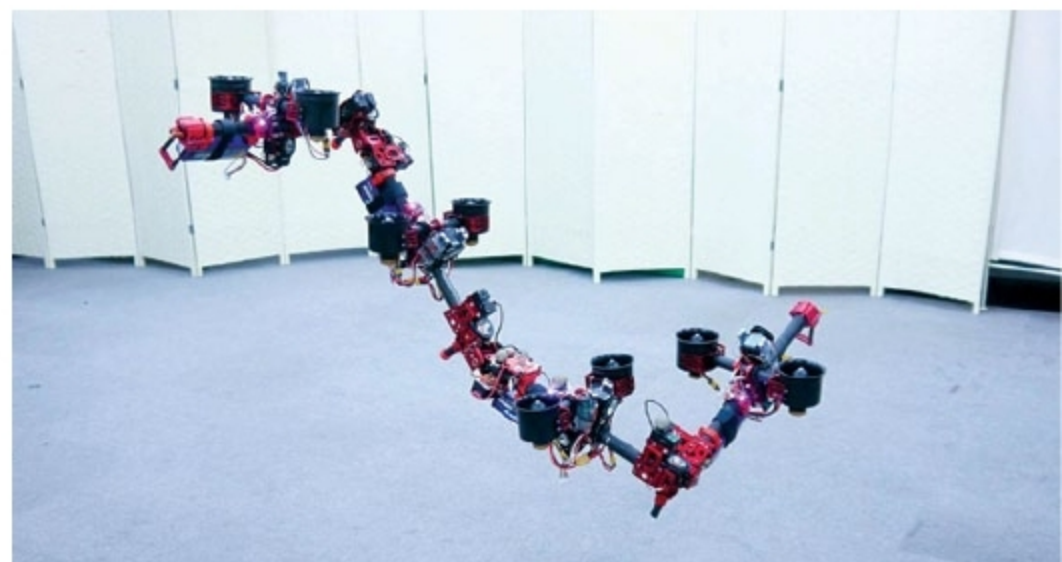
The team used spray technique to deposit thin, alternating layers of MnO_2 -coated carbon nanotubes (CNTs) and reduced graphene oxide (rGO). The layers were stacked on top of a stainless-steel mask mounted on a standard PET plastic sheet. This type of patterning not only increased the surface area, but also positioned the materials strategically for charges to move efficiently.

The layered hybrid supercapacitor showed a much larger capacitance, compared to structures that had only CNT, only rGO or a random mix of the two materials. For the same size, it also showed greater storage capacity than existing supercapacitors.

Shape-shifting robot that can move through complex environments

Roboticians at JSK Lab at University of Tokyo have developed a robot called DRAGON, which stands for Dual-rotor embedded multilink Robot with the Ability of multi-deGREE-of-freedom aerial transformatiON. As reported by IEEE, it is a modular flying robot powered by ducted fans that can transform literally on-the-fly, from a square to a snake to anything in between. This allows DRAGON to stretch out to pass through small holes and then make any other shape you want, once it is on the other side.

DRAGON is made of a series of linked modules, each consisting a pair of ducted fan thrusters that can be actuated in roll-and-pitch to vector thrust in any direction. Modules are



DRAGON can autonomously decide how to transform when given the constraints of the space it needs to pass through (Credit: JSK Lab, University of Tokyo)